Exact Refocusing of Diagonal Crosstalk for Shor Order Finding on a 10 × 12 Square-Lattice Processor: Analysis and Simulation

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Abstract

On a bipartite square-lattice register whose idle drift is diagonal (ZZ couplings plus static and quasi-static Z detunings) and whose incoherent noise is global depolarising with survival λ per circuit block, a four-segment Walsh echo in each idle window makes Shor order finding for N = 15, a = 7 output exactly λ^B p_ideal + (1 − λ^B) u, for every value of the couplings and detunings. With B = 5 blocks and λ = 0.94, the per-shot factoring success is 0.40021; no refocusing that restores the ideal unitary can raise it, because the depolarising factor commutes with every unitary layer.

GHZ population measurements identify λ exactly, a conditional Ramsey protocol identifies every coupling J_ij, and a monotone readout map recovers p_ideal without changing per-shot success. Sections 2 to 7 prove these statements. Section 8 reports an 8-qubit density-matrix simulation that reproduces the predicted success to four decimals and measures the effect of exchange coupling, which lies outside the hypotheses. Section 9 runs Layers 1a and 2 on 16- to 61-qubit Nighthawk registers split by Qrack's ACE simulator: off the seams the echo restores F_GHZ = 1.000, on them every Layer 2 mode lowers it.

1. Setting

The stack has three layers: Layer 1 calibrates the drift and the noise (§3), Layer 2 refocuses each idle window (§4), and Layer 3 runs the order-finding circuit (§6). This section fixes the objects the proofs use.

Definition 1 (lattice and register). Γ = (V, E) is the 10 × 12 square grid, vertex (r, c) adjacent to (r ± 1, c) and (r, c ± 1). The register Π is the 2 × 4 sub-grid on rows 3–4 and columns 4–7, with n = 8 qubits and coupling set E_Π. It splits as a counting register R_c (the four qubits of row 3) and a work register R_w (row 4), so d = 2^8 and d_c = 2^4.

Definition 2 (drift). During an idle window of length T the register evolves under H = H_D + H_X with

> $H_D = \Sigma_{(i,j)\in E} J_{ij} Z_i Z_j + \Sigma_i (\delta\omega_i + \varepsilon_i) Z_i, \qquad H_X = \eta \Sigma_{(i,j)\in E} J_{ij} (X_i X_j + Y_i Y_j)$

The J_{ij} and $\delta\omega_i$ are fixed real numbers. The vector ε is quasi-static: constant within one shot and drawn from a distribution μ between shots.

Definition 3 (depolarising channel). $D_p(\rho) = (1 - p)\,\rho + p\,I/d$, with survival $\lambda = 1 - p$.

Definition 4 (block). A block applies an exact gate G, an idle unitary V, then D_p:

> $\mathcal{E}_{G,V}(\rho) = D_p(V G \rho G^\dagger V^\dagger)$

A circuit is B blocks in sequence; its output state is averaged over μ.

Definition 5 (Walsh sequence). Let A and B be the sublattices of Lemma 1, let X_A and X_B be the products of X over A and over B, and let $\tau = T/4$. The refocused idle unitary is

> $W_T(H) = X_A \cdot \exp(-i\tau H) \cdot X_B \cdot \exp(-i\tau H) \cdot X_A \cdot \exp(-i\tau H) \cdot X_B \cdot \exp(-i\tau H)$

Definition 6 (compensation). Given a diagonal estimate $\hat{H} = \Sigma \hat{J}_{ij} Z_i Z_j + \Sigma \delta\hat{\omega}_i Z_i$, the compensated idle unitary is $\exp(+iT\hat{H}) \exp(-iTH)$.

Definition 7 (per-shot success). $S \subset \{0, \ldots, 15\}$ is the set of counting outcomes y from which continued fractions on $y/16$ return a non-trivial factor pair. For a counting distribution p, $P(p) = \Sigma_{y\in S} p(y)$.

Definition 8 (OTOC). For Hermitian unitary operators W, V and idle unitary U, with $W(t) = (U^\dagger)^t\, W\, U^t$:

> $C(t) = 2^{-n} \cdot \|[W(t), V]\|^2$, with $\|\cdot\|$ the Frobenius norm

Assumptions.

- A1. Echo pulses are instantaneous and exact.
- A2. Incoherent noise in a block is the global depolarising channel D_p, applied after the idle window.
- A3. Crosstalk acting during a gate is absorbed into the adjacent idle window; gates G are exact.
- A4. $\eta = 0$ (no exchange) wherever a statement is called exact; $\eta > 0$ is treated numerically in §8.

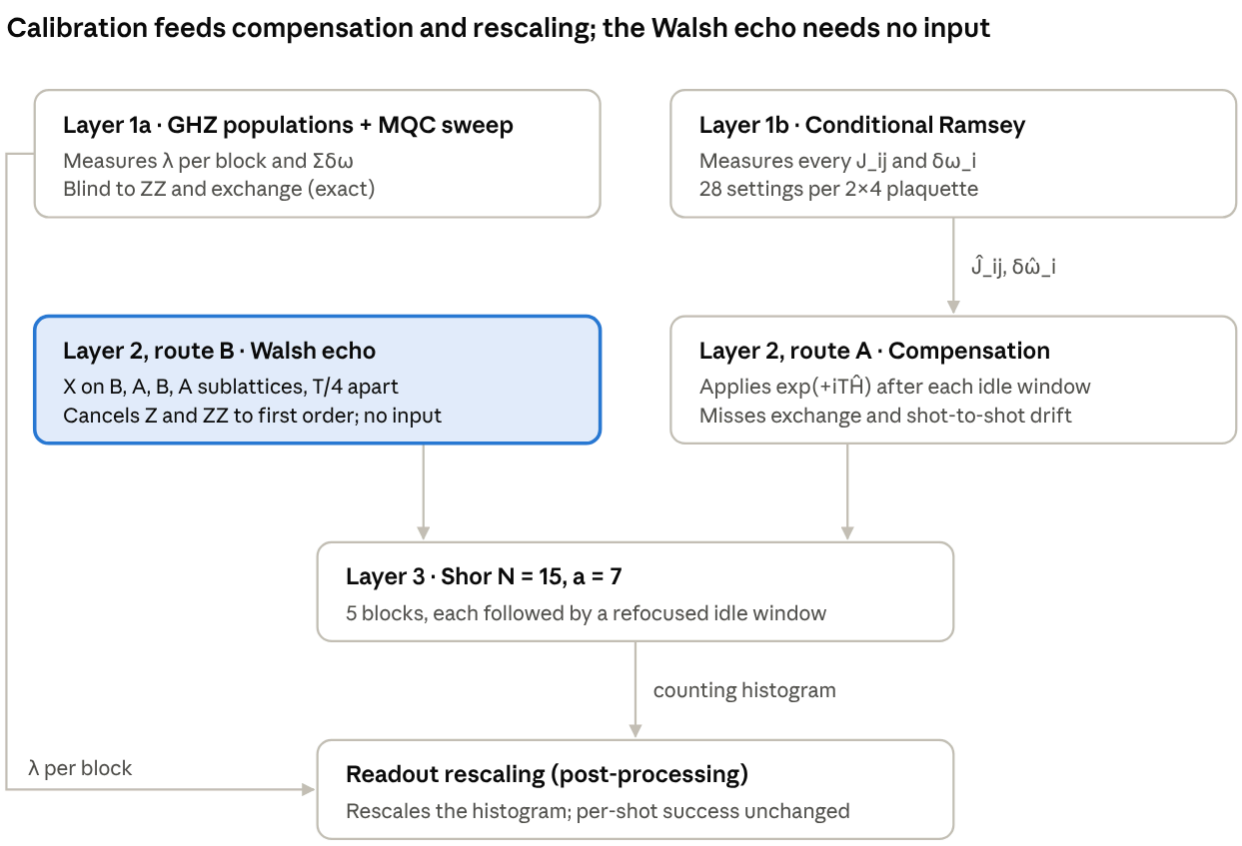


Figure 1 · the three-layer stack: calibration (§3), refocusing (§4), payload (§6), post-processing (§7)

Route B is the refocusing that §4 proves exact; route A is analysed for comparison.

2. Lattice

Lemma 1. The 10 × 12 square grid has 218 edges and is bipartite under the colouring A = {(r, c) : r + c even}, B = {(r, c) : r + c odd}. Its maximum degree is 4 and its mean degree is 436/120 ≈ 3.633. The same holds for every rectangular sub-grid, in particular for Π.

Proof. Each of the 10 rows has 11 horizontal edges and each of the 12 columns has 9 vertical edges, so |E| = 10·11 + 9·12 = 110 + 108 = 218. Every edge changes r + c by exactly 1, so it joins A to B. An interior vertex has the four neighbours (r ± 1, c), (r, c ± 1), and no vertex has more. The degree sum is 2|E| = 436 over 120 vertices. A sub-grid inherits the colouring. ∎

The counts match the published Nighthawk layout: 120 qubits joined by 218 tunable couplers in a square lattice (Help Net Security). For Π, |E_Π| = 2·3 + 4 = 10.

3. Calibration (Layer 1)

The GHZ state identifies the depolarising survival λ and the detuning sum exactly, and is provably blind to every ZZ coupling; a conditional Ramsey protocol identifies each J_ij exactly. Write |0⟩ for the n-qubit string 0…0 and |1⟩ for 1…1 where no confusion arises.

Lemma 2 (GHZ endpoints are eigenvectors). For all J, δω, ε and η, $H|0\ldots0\rangle = E_0|0\ldots0\rangle$ and $H|1\ldots1\rangle = E_1|1\ldots1\rangle$, with

> $E_0 = \Sigma_{(i,j)} J_{ij} + \Sigma_i (\delta\omega_i + \varepsilon_i)$, $E_1 = \Sigma_{(i,j)} J_{ij} - \Sigma_i (\delta\omega_i + \varepsilon_i)$, $E_0 - E_1 = 2\,\Sigma_i (\delta\omega_i + \varepsilon_i)$

Proof. Z_i and Z_iZ_j are diagonal, with $Z_iZ_j = +1$ on both strings and $Z_i = +1$ on 0…0, −1 on 1…1. On a pair of equal bits, $X_iX_j|00\rangle = |11\rangle$ and $Y_iY_j|00\rangle = (i)(i)|11\rangle = -|11\rangle$, so $(X_iX_j + Y_iY_j)|00\rangle = 0$, and likewise on |11⟩. Summing the terms gives E_0 and E_1. ∎

Corollary 2.1 (cat states cannot see ZZ). Let $x \in \{0,1\}^n$ and $\bar{x}$ its complement. Under H_D, the relative phase between |x⟩ and |x̄⟩ does not depend on any J_{ij}. Hence no measurement on any state in span{|x⟩, |x̄⟩} after diagonal drift carries information about the couplings.

Proof. Z_iZ_j has eigenvalue $(-1)^{(x_i + x_j)}$ on |x⟩ and $(-1)^{((1-x_i) + (1-x_j))} = (-1)^{(x_i + x_j)}$ on |x̄⟩. The ZZ part of the energy is therefore equal on both strings and cancels in the phase difference; populations are invariant because H_D is diagonal. ∎

Lemma 3 (population identity). Start from $|GHZ\rangle = (|0\ldots0\rangle + |1\ldots1\rangle)/\sqrt{2}$ and apply L blocks with G = I and V = exp(−iTH) (no echo). Let P_0, P_1 be the output populations of 0…0 and 1…1. Then, for every μ,

> $P_0 + P_1 = \lambda^L + (1 - \lambda^L) \cdot 2/d$, $\lambda = [(P_0 + P_1 - 2/d) / (1 - 2/d)]^{(1/L)}$

Proof. By Lemma 2, V acts on span{|0…0⟩, |1…1⟩} as $\mathrm{diag}(\exp(-iE_0T), \exp(-iE_1T))$, so $V\rho V^\dagger$ keeps $P_0 + P_1 = 1$ for ρ in that span. Since $V\,I\,V^\dagger = I$, induction gives the L-block output $\lambda^L V^L \rho (V^\dagger)^L + (1 - \lambda^L)\, I/d$. The maximally mixed term puts 2/d on the two strings. Averaging over μ is linear and leaves the identity unchanged. Solve for λ; the map is invertible because $d > 2$. ∎

Lemma 4 (multiple-quantum-coherence readout). Let $c = \langle 0\ldots0|\rho|1\ldots1\rangle$ and $M(\varphi) = \otimes_q (\cos\varphi\, X_q + \sin\varphi\, Y_q)$. For any ρ,

> $\mathrm{Tr}[M(\varphi)\rho] = \Sigma_{(a \in \{0,1\}^n)} \exp(i\varphi(2|a| - n)) \cdot \rho_{(\bar{a},a)}$

so the coefficient of exp(inφ) is exactly c. If the only non-zero coherence of the form ρ_(ā,a) is c and its conjugate, as for every output of Lemma 3, then Tr[M(φ)ρ] = C cos(nφ + Φ) with C = 2|c|, Φ = arg c. Moreover

> ½ (P_0 + P_1 + C) = max over θ of ⟨GHZ_θ| ρ |GHZ_θ⟩, |GHZ_θ⟩ = (|0…0⟩ + exp(iθ) |1…1⟩) / √2

Proof. cos φ X + sin φ Y = exp(−iφ)|0⟩⟨1| + exp(iφ)|1⟩⟨0|, so M(φ) has non-zero entries only at (a, ā), equal to the product over qubits of exp(−iφ) where a_q = 0 and exp(+iφ) where a_q = 1, that is exp(iφ(2|a| − n)). Then Tr[Mρ] = Σ_a M_(a,ā) ρ_(ā,a). Frequency n arises only from a = 1…1, with coefficient ρ_(0…0, 1…1) = c; frequency −n gives c*. For the fidelity, ⟨GHZ_θ|ρ|GHZ_θ⟩ = ½(P_0 + P_1 + 2 Re(exp(iθ) c)), maximal at θ = −arg c. ∎

Sampling. With φ_m = πm/(n + 1), m = 0, …, 2n + 1, the 2n + 2 samples separate all frequencies in {−n, …, n} (span 2n < 2n + 2), so the discrete Fourier coefficient at n returns c exactly for any ρ.

Corollary 4.1 (detuning sum). If μ is symmetric with a positive characteristic function (for example Gaussian), the output of Lemma 3 has arg c = −2LT Σ_i δω_i (mod 2π), so Σ δω_i = −Φ/(2LT) whenever 2LT |Σ δω_i| < π.

Proof. By Lemmas 2 and 3, c = (λ^L/2) exp(−2iLTΣδω) E_μ[exp(−2iLTΣε)], and the expectation is real and positive by hypothesis. ∎

Lemma 5 (conditional Ramsey, η = 0). Prepare qubit i in |+⟩ and every other qubit k in a computational state with sign z_k = +1 for |0⟩, −1 for |1⟩. After time τ under H_D,

> ⟨X_i⟩ + i⟨Y_i⟩ = exp(2iτ h_i(z)) · E_μ[exp(2iτ ε_i)], h_i(z) = δω_i + Σ_(k∼i) J_ik z_k

Under the hypothesis of Corollary 4.1 and 2τ|h_i| < π, the estimators

> $\hat{J}$_ij = ½ [h_i(z_j = +1) − h_i(z_j = −1)], δ$\hat{\omega}$_i = h_i(+1, …, +1) − Σ_(k∼i) $\hat{J}$_ik

(other z_k = +1) return J_ij and δω_i exactly.

Proof. H_D is diagonal, so the product state stays a product and qubit i sees the field h_i(z) + ε_i: |0⟩ ↦ exp(−iτ(h+ε))|0⟩, |1⟩ ↦ exp(iτ(h+ε))|1⟩. For a|0⟩ + b|1⟩, ⟨X⟩ + i⟨Y⟩ = 2a*b = exp(2iτ(h+ε)). Averaging over μ multiplies by a positive real factor, so the argument is 2τh_i. Flipping z_j changes h_i by −2J_ij. ∎

One register needs n + 2|$E_Π$| = 28 settings when each coupling is read from both ends. With η > 0 the product state is no longer preserved and the estimators acquire an O(η) bias, measured in §8.

4. Refocusing (Layer 2)

The Walsh sequence of Definition 5 cancels every diagonal drift term exactly, to all orders and for any unknown quasi-static detuning; compensation cancels only what was measured.

Theorem 1 (exact Walsh refocusing). Assume A1 and $\eta = 0$. For all couplings J, detunings $\delta\omega$, realisations ε and window lengths T,

> $W_T(H_D) = I$

Proof. Write $U = \exp(-i\tau H_D)$ and $Q_1 = X_B$, $Q_2 = X_A$, $Q_3 = X_B$, $Q_4 = X_A$, so $W_T = Q_4 U Q_3 U Q_2 U Q_1 U$. Define the frames $F_0 = I$ and $F_k = Q_k \cdots Q_1$. Since X_A and X_B have disjoint supports, commute and square to I, $F_4 = I$. Because $F_k F_{k-1}^\dagger = Q_k$,

> $W_T = F_4 (F_3^\dagger U F_3)(F_2^\dagger U F_2)(F_1^\dagger U F_1)(F_0^\dagger U F_0) = \exp(-i\tau H_3) \exp(-i\tau H_2) \exp(-i\tau H_1) \exp(-i\tau H_0), \qquad H_k = F_k^\dagger H_D F_k$

The frames are $F_0 = I$, $F_1 = X_B$, $F_2 = X_A X_B$, $F_3 = X_A$. Conjugation by X flips the sign of Z on that qubit, and by Lemma 1 every coupling joins A to B, so

> $H_k = \sum_{(i,j)} a_k b_k J_{ij} Z_i Z_j + \sum_{i\in A} a_k (\delta\omega_i + \varepsilon_i) Z_i + \sum_{i\in B} b_k (\delta\omega_i + \varepsilon_i) Z_i, \qquad a = (+,+,-,-), \quad b = (+,-,-,+)$

All H_k are diagonal and therefore commute, so the product equals $\exp(-i\tau \sum_k H_k)$. Since $\sum a_k = \sum b_k = \sum a_k b_k = 0$, $\sum_k H_k = 0$ and $W_T = I$. ∎

Remark on Theorem 1. No condition on $T\|H\|$ is needed: the toggled Hamiltonians commute, so there are no higher-order terms. With $\eta > 0$, $X_i X_j$ is invariant under every frame and $Y_i Y_j$ picks up the sign $a_k b_k$, so the first-order average Hamiltonian is $\tfrac{1}{4} \sum_k H_k = \eta \sum J_{ij} X_i X_j$, plus commutator terms of order $\tau^2\|H\|^2$. The echo therefore leaves an XX residual.

Proposition 1 (compensation residual). For diagonal H_D and $\hat{H}$,

> $\exp(iT\hat{H}) \exp(-iTH_D) = \exp(-iT [\sum_{(i,j)} (J_{ij} - \hat{J}_{ij}) Z_i Z_j + \sum_i (\delta\omega_i - \delta\hat{\omega}_i + \varepsilon_i) Z_i])$

This is I for every realisation only if $\hat{J} = J$, $\delta\hat{\omega} = \delta\omega$ and $\varepsilon = 0$ almost surely. If the ε_i are independent $N(0, \sigma^2)$, each window multiplies the coherence between two strings that differ only at qubit i by $\exp(-2T^2\sigma^2)$, in addition to the deterministic residual phase.

Proof. The exponents commute, so they add. The ε-dependent phase difference between the two strings is $2T\varepsilon_i$, and $\mathbb{E}[\exp(2iT\varepsilon_i)] = \exp(-2T^2\sigma^2)$ for a centred Gaussian. ∎

With $T = 4.4$ and $\sigma = 0.015$ this factor is 0.9913 per window and per qubit, which the echo of Theorem 1 removes and compensation cannot.

Lemma 6 (OTOC identity). For Hermitian unitary W and V, with $F(t) = 2^{-n}\,\mathrm{Tr}[W(t)\,V\,W(t)\,V]$,

> $$C(t) = 2\,(1 - F(t)), \qquad F(t) \text{ is real}, \qquad 0 \leqslant C(t) \leqslant 4$$

Proof. $W(t)$ is again Hermitian and unitary. $[W(t), V]^\dagger[W(t), V] = (VW - WV)(WV - VW) = 2I - VWVW - WVWV$, with $W = W(t)$. By cyclicity $\mathrm{Tr}[VWVW] = \mathrm{Tr}[WVWV]$, and that trace equals its own conjugate $\mathrm{Tr}[(WVWV)^\dagger] = \mathrm{Tr}[VWVW]$, so F is real. $WVWV$ is unitary, so $|F| \leqslant 1$. ∎

Lemma 7 (short-time growth). If $[W, V] = 0$ and $U = \exp(-itH)$, then

> $$C(t) = t^2 \cdot 2^{-n} \cdot \|[[H, W], V]\|^2 + O(t^3)$$

Proof. $W(t) = W + it[H, W] + O(t^2)$, so $[W(t), V] = it[[H, W], V] + O(t^2)$; take the squared Frobenius norm. ∎

Corollary 7.1 (refocused OTOC). Under the hypotheses of Theorem 1, with one Walsh window per time step, $C(t) = C(0)$ for all t; in particular $C \equiv 0$ when $[W, V] = 0$.

Proof. Each step is $W_T(H_D) = I$, so $W(t) = W$. ∎

The OTOC measures coherent operator spreading only. It is a diagnostic of Layer 2, not a protection mechanism, and it is insensitive to the depolarising part of the channel, which §5 treats.

5. Depolarising invariance

The survival factor λ^B is fixed by the noise alone: every unitary layer, refocusing included, commutes with it and can only change the noiseless part of the output.

Lemma 8 (commutation and composition). For every unitary U and $p, q \in [0, 1]$,

> $$U\,D_p(\rho)\,U^\dagger = D_p(U\,\rho\,U^\dagger), \qquad D_p \circ D_q = D_r \text{ with } r = 1 - (1 - p)(1 - q)$$

Proof. $UIU^\dagger = I$ gives the first identity. For the second, $(1 - p)[(1 - q)\rho + qI/d] + pI/d = (1 - p)(1 - q)\rho + [1 - (1 - p)(1 - q)]\,I/d$. ∎

Theorem 2 (fixed depolarising factor). For any gates $G_1, \ldots, G_B$ and any idle unitaries $V_1, \ldots, V_B$, which may depend on the realisation ε, the B-block output averaged over μ is

$$\rho_{out} = \lambda^B \cdot E_\mu[U_V \rho_0 U_V^\dagger] + (1 - \lambda^B) \cdot I/d, \qquad U_V = V_B G_B \cdots V_1 G_1$$

The counting marginal is therefore $p = \lambda^B p_V + (1 - \lambda^B) \upsilon$, with υ uniform on the d_c outcomes, and

$$P(p) = \lambda^B P(p_V) + (1 - \lambda^B) \cdot |S| / d_c$$

Proof. By Lemma 8 each block's D_p can be moved past all later unitaries, and the B copies compose to $D_{(1-\lambda^B)}$. Averaging over μ is linear. The partial trace of I/d over R_w is I/d_c. P is linear in p. ∎

Corollary of Theorem 2. No choice of idle unitaries changes λ^B. A refocusing that restores the ideal circuit for every ε, $U_V = U_{ideal}$, attains $P = \lambda^B P(p_{ideal}) + (1 - \lambda^B) |S|/d_c$, and this is the value of every such refocusing.

6. Shor payload (Layer 3)

For N = 15 and a = 7 the ideal counting distribution is uniform on {0, 4, 8, 12}, and exactly two of the sixteen outcomes factor N, so one ideal shot succeeds with probability 1/2.

Definition 9 (controlled multipliers). For k = 0, …, 3 let $m_k = 7^{(2^k)} \bmod 15$ and

$$c\text{-}U_k : |c\rangle|y\rangle \mapsto |c\rangle|m_k\, y \bmod 15\rangle \text{ if } c_k = 1 \text{ and } 0 < y < 15, \text{ and } |c\rangle|y\rangle \text{ otherwise}$$

where c_k is bit k of the counting value c. Each $c\text{-}U_k$ is a permutation of basis states, hence unitary, because $\gcd(m_k, 15) = 1$ makes $y \mapsto m_k\, y \bmod 15$ a bijection of $\{1, \ldots, 14\}$. The circuit is B = 5 blocks: $c\text{-}U_0, \ldots, c\text{-}U_3$, then the inverse QFT on R_c, from the input $|+\rangle^{\otimes 4} \otimes |1\rangle$.

Lemma 9 (ideal counting distribution). $p_{ideal}(y) = 1/4$ for $y \in \{0, 4, 8, 12\}$ and 0 otherwise.

Proof. The four multipliers compose to $|c\rangle|1\rangle \mapsto |c\rangle|7^c \bmod 15\rangle$. The multiplicative order of 7 mod 15 is 4 (7, 4, 13, 1), so writing $c = 4j + s$,

$$\tfrac{1}{4} \Sigma_{c=0\ldots15} |c\rangle|7^c\rangle = \tfrac{1}{4} \Sigma_{s=0\ldots3} \left(\Sigma_{j=0\ldots3} |4j + s\rangle\right) |7^s\rangle$$

The inverse QFT sends $|x\rangle$ to $\tfrac{1}{4} \Sigma_y \exp(-2\pi i x y/16)|y\rangle$, so the amplitude on $|y\rangle|7^s\rangle$ is $(1/16) \exp(-2\pi i s y/16) \Sigma_j \exp(-2\pi i j y/4)$. The inner sum is 4 if 4 divides y and 0 otherwise. Hence $p(y) = \Sigma_s (1/4)^2 = 1/4$ for $y \equiv 0 \bmod 4$. ∎

Lemma 10 (success set). With continued fractions taken to denominators at most 15, S = {4, 12}, so |S| = 2 and P(p_ideal) = 1/2.

Proof. By enumeration of all sixteen outcomes. A candidate period r is accepted when r is even, $7^r \equiv 1$ and x = 7^(r/2) mod 15 ∉ {1, 14}; then gcd(x ± 1, 15) is the factor pair.

y	Convergent	r	7^r mod 15	x = 7^(r/2) mod 15	In S
0	0	1	7	—	no
4, 12	1/4, 3/4	4	1	4	yes: gcd(3, 15) = 3, gcd(5, 15) = 5
8	1/2	2	4	—	no
2, 6, 10, 14	1/8, 3/8, 5/8, 7/8	8	1	1	no
1, 15	1/15, 14/15	15	13	—	no
3, 13	2/11, 9/11	11	13	—	no
5, 11	4/13, 9/13	13	7	—	no
7, 9	4/9, 5/9	9	7	—	no

Only y = 4 and 12 pass, and each carries ideal probability 1/4. ∎

7. Main result

Theorem 3. Assume A1–A4 with η = 0. Run the circuit of §6 on Π with every idle window replaced by the Walsh sequence W_T(H) of Definition 5. Then for all couplings J, detunings δω, window lengths T and distributions μ,

> $p = \lambda^5\, p_{ideal} + (1 - \lambda^5)\, u, \qquad P(p) = \lambda^5/2 + (1 - \lambda^5)/8$

For λ = 0.94, $\lambda^5 = 0.733904$ and P = 0.366952 + 0.033262 = 0.400214.

Proof. Lemma 1 gives the bipartition that Theorem 1 needs, so every idle window equals I for every ε, and U_V = U_ideal. Theorem 2 with B = 5 gives $p = \lambda^5\, p_{ideal} + (1 - \lambda^5)\, u$. Lemmas 9 and 10 give P(p_ideal) = 1/2 and |S|/d_c = 2/16. Substitute. ∎

By the corollary of Theorem 2, 0.400214 is the value of every refocusing that restores the ideal circuit; refocusing cannot exceed it by lowering the depolarising contribution.

Corollary A (calibration transfer). When the calibration blocks of Lemma 3 use the same D_p as the circuit (A2), the survival λ^5 needed in Theorem 3 is identified from GHZ populations alone: $\Lambda = [(P_0 + P_1 - 2/d)/(1 - 2/d)]^{(5/L)}$.

Proof. Lemma 3 identifies the per-block λ; Theorem 2 raises it to the number of blocks. ∎

Corollary B (readout rescaling). Let Π_Δ be Euclidean projection onto the probability simplex and

$$R_\Lambda(p) = \Pi_\Delta[(p - (1 - \Lambda)\,\upsilon) / \Lambda]$$

1. Under Theorem 3 with $\Lambda = \lambda^5$, $R_\Lambda(p) = p_{ideal}$ exactly.
2. For every p, R_Λ preserves the weak order of the outcome probabilities, so it cannot promote an outcome over another.
3. R_Λ acts on an estimated histogram, not on device samples, so the per-shot success $P(p)$ of the device is unchanged by it.
4. Each rescaled entry has variance $\mathrm{Var}(\hat{p}(y))/\Lambda^2$, a factor $1/\Lambda^2 \approx 1.857$ at $\Lambda = 0.733904$.

Proof. (1) The argument of Π_Δ equals p_{ideal}, which lies in Δ and is fixed by Π_Δ. (2) The affine map $v \mapsto (v - (1 - \Lambda)/d_c)/\Lambda$ is strictly increasing in each coordinate, and Euclidean projection onto Δ has the form $v \mapsto \max(v - \theta, 0)$ for one scalar θ, which is non-decreasing. (3) is immediate. (4) Scaling a random variable by $1/\Lambda$ scales its variance by $1/\Lambda^2$. ∎

8. Numerical corroboration

An 8-qubit density-matrix simulation of Π returns per-shot success 0.4002 for the Walsh echo whenever the drift is diagonal, matching Theorem 3′s 0.400214 to four decimals; adding exchange, which the theorem excludes, lowers it to 0.3912.

Model parameter	Value
J_{ij}	$0.065 \times U(0.75, 1.25)$ rad per time unit
Exchange η	0.15 (0 where stated)
$\delta\omega_q$	$0.022 \times ((q \bmod 3) - 1)$
Quasi-static ε_q	$N(0, 0.015^2)$ per shot (0 where stated)
Depolarising p	0.06 per block, all modes
Window T, Ramsey wait τ, GHZ blocks L	4.4, 2.0, 3
Shots per setting	4000
Noise realisations	256 per calibration, 32 per circuit run

Calibration (Lemmas 2–5), default parameters:

Quantity	Estimate	True	Statement
GHZ relative phase from ZZ, 10 units	0.000 rad	0	Lemma 2
GHZ relative phase from XX + YY, 10 units	0.000 rad	0	Lemma 2
GHZ relative phase from Z detuning, 10 units	0.440 rad	0.440 rad	Lemma 2
λ per block	0.9401	0.9400	Lemma 3
Σ δω_i	−0.0227	−0.0220	Corollary 4.1
Largest relative error in Ĵ_ij	5.0 %	0 at η = 0	Lemma 5
Λ = λ⁵	0.7344	0.7339	Corollary A

Per-shot success by noise content. The first two rows satisfy the hypotheses of Theorem 3; the last two do not.

Drift terms present	Theorem 3 applies	Unmitigated	Compensated	Walsh echo
Z + ZZ only	yes	0.098	0.393	0.4002
Z + ZZ + quasi-static ε	yes	0.099	0.353	0.4002
Z + ZZ + exchange	no	0.085	0.337	0.391
All terms (default)	no	0.088	0.305	0.391

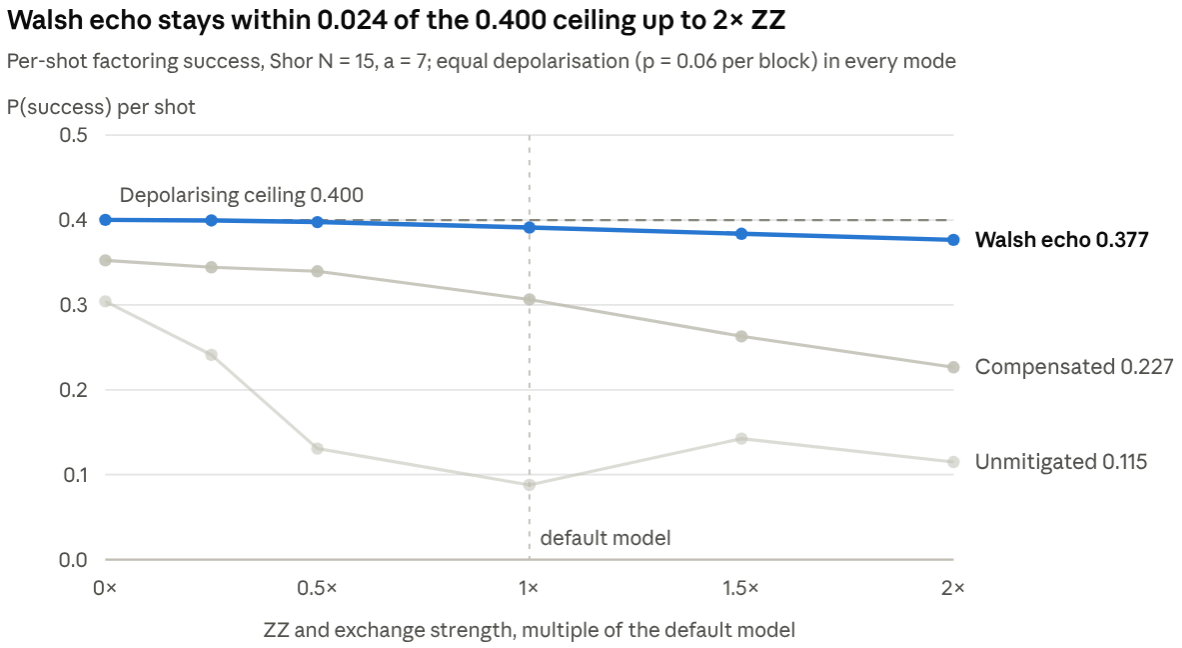
The echo is insensitive to ε, as Theorem 1 states; compensation loses 0.007 to estimation error and a further 0.040 to ε, as Proposition 1 predicts. Under the hypotheses, R_Λ returns fidelity 0.9995 to p_ideal (Corollary B). With all terms the fidelities to p_ideal are 0.792 (echo), 0.626 (compensated) and 0.197 (unmitigated); after R_Λ they read 0.988, 0.762 and 0.177, with per-shot success unchanged.

OTOC after t idle windows (W = X on the first qubit, V = X on the second, ε = 0, η = 0.15):

Pair	Mode	t = 1	t = 3	t = 5
Neighbours (1, 5)	Unmitigated	1.371	3.189	1.053
Neighbours (1, 5)	Compensated	0.037	0.305	0.730
Neighbours (1, 5)	Walsh echo	0.001	0.007	0.018
Distance 4 (0, 7)	Unmitigated	0.000	0.005	0.079
Distance 4 (0, 7)	Compensated	0.000	0.000	0.003

Pair	Mode	t = 1	t = 3	t = 5
Distance 4 (0, 7)	Walsh echo	0.000	0.000	0.000

The echo's non-zero neighbour values come from the XX residual of the Remark on Theorem 1; at η = 0 Corollary 7.1 makes them exactly zero. The unmitigated oscillation reflects the short recurrence time of 8 qubits.



shrs_nighthawk_stealth.py sweep, seed 42 · 6 strengths × 3 modes · 32 noise realisations per point · η = 0.15 throughout

Scaling the ZZ and exchange strengths together moves the echo only from 0.4002 to 0.3766 over 0 to 2×, all of it due to exchange. The full default run takes 2 min 35 s on one CPU core.

9. Chip-scale implementation

nighthawk_ghz.py carries Layers 1a and 2 from the 8-qubit register Π to whole Nighthawk-class registers of 16 to 61 qubits, simulated with Qrack's ACE backend. Off the seams it reproduces §3 and §4 exactly; on the seams no Layer 2 mode helps.

ACE splits a register into patches, each held by its own simulator. A seam qubit belongs to more than one patch, and a coupler between patches (a seam CX) is applied through replicas rather than as an exact two-qubit gate. Every run is checked against an exact reference (QrackSimulator) with the same gates and the same noise draws.

Object in the proof	Implementation in nighthawk_ghz.py
Register Π (Definition 1)	3-qubit chains tiled over the register (rows and columns, offsets 0 to 2), or one GHZ over the whole register along a spanning tree: bfs, or seam (fewest seam crossings)
Drift H_D (Definition 2, η = 0)	Per idle window: Z rotations for δω + ε; each coupling inside a unit as CX · Rz · CX; each coupling to an idle neighbour as a Z rotation conditioned on that neighbour
Walsh sequence W_T (Definition 5)	X on sublattice B, A, B, A between four segments of τ = T/4; units placed so that no two share a coupler
Compensation (Definition 6)	Oracle $\hat{H}$ = H_D without ε
λ (Lemma 3)	From $P_0 + P_1$ per unit
MQC readout (Lemma 4)	2k + 2 phases $\varphi_m = \pi m/(k+1)$ on a k-qubit unit; $C = 2\lvert c\rvert$, $F_{GHZ} = (P_0 + P_1 + C)/2$; used up to 12 qubits
Fidelity above 12 qubits	Clifford witness $F \geqslant P_0 + P_1 + P(\text{even X parity}) - 1$
§8 noise model	J = 0.065 × U(0.75, 1.25), $\delta\omega_q = 0.022\,((q \bmod 3) - 1)$, σ = 0.015, T = 4.4, L = 3, seed 42; K = 4 noise realisations per setting (256 in §8)

The registers are nighthawk2p16, nighthawk2p28 and nighthawk2p57 (two patches each) and nighthawk61r2 (61 qubits, two patches, 6 seam qubits). The release lattice is stored as 12 rows × 10 columns, the transpose of Definition 1; Lemma 1 is unchanged. All runs use PyQrack 2.29.5 with Qrack 10.13.2 on CPU.

9.1 Layer 1a across the seams

Without drift, a 3-qubit GHZ chain keeps $F_{GHZ} = 1.000$ everywhere except on chains that hold a seam qubit, where it falls to 0.77 with one seam CX and to 0.35–0.43 with two.

Register	Units	Off the seam	Holding a seam qubit, 1 seam CX	Holding a seam qubit, 2 seam CXs
nighthawk2p28 (3 seam qubits)	30	1.002 (16 units)	0.767 (9)	0.353 (5)
nighthawk61r2 (6 seam qubits)	82	1.000 (58)	0.773 (15)	0.426 (9)

Mean F_GHZ from Lemma 4; 256 shots × 4 backends per setting on nighthawk61r2, × 2 on nighthawk2p28. The exact reference reads 0.996–1.002 in every row. Lemma 3 gives λ per block 1.000, 0.80–0.83 and 0.55–0.62 for the three columns. Values above 1 are shot noise in C.

- **The loss sits on the seam qubits, not around them.** Every chain that only touches a seam qubit's neighbour is at 1.000 with phase 0.00 ± 0.01.
- **It is a decorrelation at the crossing, not an error per gate.** Folding each seam CX three times (fold:1) moves the seam units only from 0.638 to 0.622 in F_0 and leaves every other unit at 1.000.

9.2 Layer 2 across the seams

Off the seam the Walsh echo restores F_GHZ = 1.001, as Theorem 1 predicts; on chains holding a seam qubit every Layer 2 mode lowers the fidelity below the no-drift value, and the exact reference stays at 0.88–1.00.

nighthawk61r2 units	No drift	Unmitigated	Compensated (oracle $\hat{H}$)	Walsh echo	Exact: unmitigated / compensated / echo
Off the seam (58)	1.000	0.901 (F_0 0.581)	0.900 (F_0 0.877)	1.001	0.901 / 0.900 / 1.002
Holding a seam qubit, 1 seam CX (15)	0.773	0.650	0.627	0.671	0.880 / 0.884 / 0.999
Holding a seam qubit, 2 seam CXs (9)	0.426	0.304	0.274	0.240	0.886 / 0.887 / 1.000

Mean F_GHZ, L = 3 windows, 256 shots × K = 4 realisations per setting, the same ε draws for ACE and the exact reference. nighthawk2p28 repeats the pattern: 1.000 off the seam under the echo, 0.183 on the two-seam-CX units.

- **Off the seam, ACE reproduces §4.** Unmitigated drift leaves F_GHZ at 0.90 (the ε dephasing of Proposition 1 over K = 4 draws) and F_0 at 0.58 (the deterministic phase). Compensation restores the phase (F_0 0.88) but not ε. The echo removes both, phase 0.00 ± 0.01 (Theorem 1, Corollary 7.1).

- **On the seam, the drift itself crosses the seam.** A coupling J_ij between two patches is one more seam crossing per segment, so the echo's four segments cost most: on the two-seam-CX units it falls to 0.240, below unmitigated (0.304).
- **Corollary 4.1 needs the calibration's K.** At K = 4 the ε phase spread over LT = 13.2 is 2LTσ√k ≈ 0.7 rad on a 3-qubit unit, larger than the δω signal; the detuning-sum estimate needs K ≈ 256, as in §8.

9.3 One GHZ over the whole register

A single GHZ over the register does not survive the seam on ACE: the coherence C and ⟨X…X⟩ are ≈ 0 in every chip-wide run, while the exact reference is ideal.

Register	Tree	Depth	CXs exact / seam	$P_0 + P_1$	Fidelity	Seam CXs that break
nighthawk2p16, no drift	bfs	5	10 / 5	0.583	F_GHZ 0.293	—
nighthawk2p16, Walsh echo	bfs	5	10 / 5	0.000	—	—
nighthawk2p16, no drift	seam	6	11 / 4	0.456	F_GHZ 0.238	—
nighthawk2p28, no drift	bfs	5	21 / 6	0.293	F ⩾ −0.19	3 of 6
nighthawk2p28, no drift	seam	6	23 / 4	0.513	F ⩾ 0.02	2 of 4
nighthawk2p57, no drift	bfs	7	44 / 12	0.267, then 0.000	F ⩾ −0.24	4 of 12
nighthawk2p57, no drift	seam	10	50 / 6	0.000, then 0.005	F ⩾ −0.51	3 of 6

nighthawk2p16 uses Lemma 4 (256 shots × 4 backends); the larger registers use the Clifford witness (512 shots × 2). A seam CX that breaks does so with probability 0.45–0.52 and takes its whole subtree with it; the other seam CXs break at 0.000–0.010, and no CX inside one patch breaks. The two trees differ less than repeated runs do, so fewer seam crossings are not shown to help. nighthawk61r2 has no chip-wide result on this build: its stabilizer patches fail in ACE's sampler.

Remark (the seam lies outside A2). In the same run, units inside one patch keep $P_0 + P_1 = 1.000$ while units holding a seam qubit lose it, so the seam acts as a channel on part of the register, not as the global D_p of Definition 3. Lemma 8 needs a channel that commutes with every unitary; a channel acting on one side of the cut does not commute with gates that cross it. Theorems 2 and 3 therefore say nothing about seam units, and §9.2 shows that refocusing there can lower the fidelity.

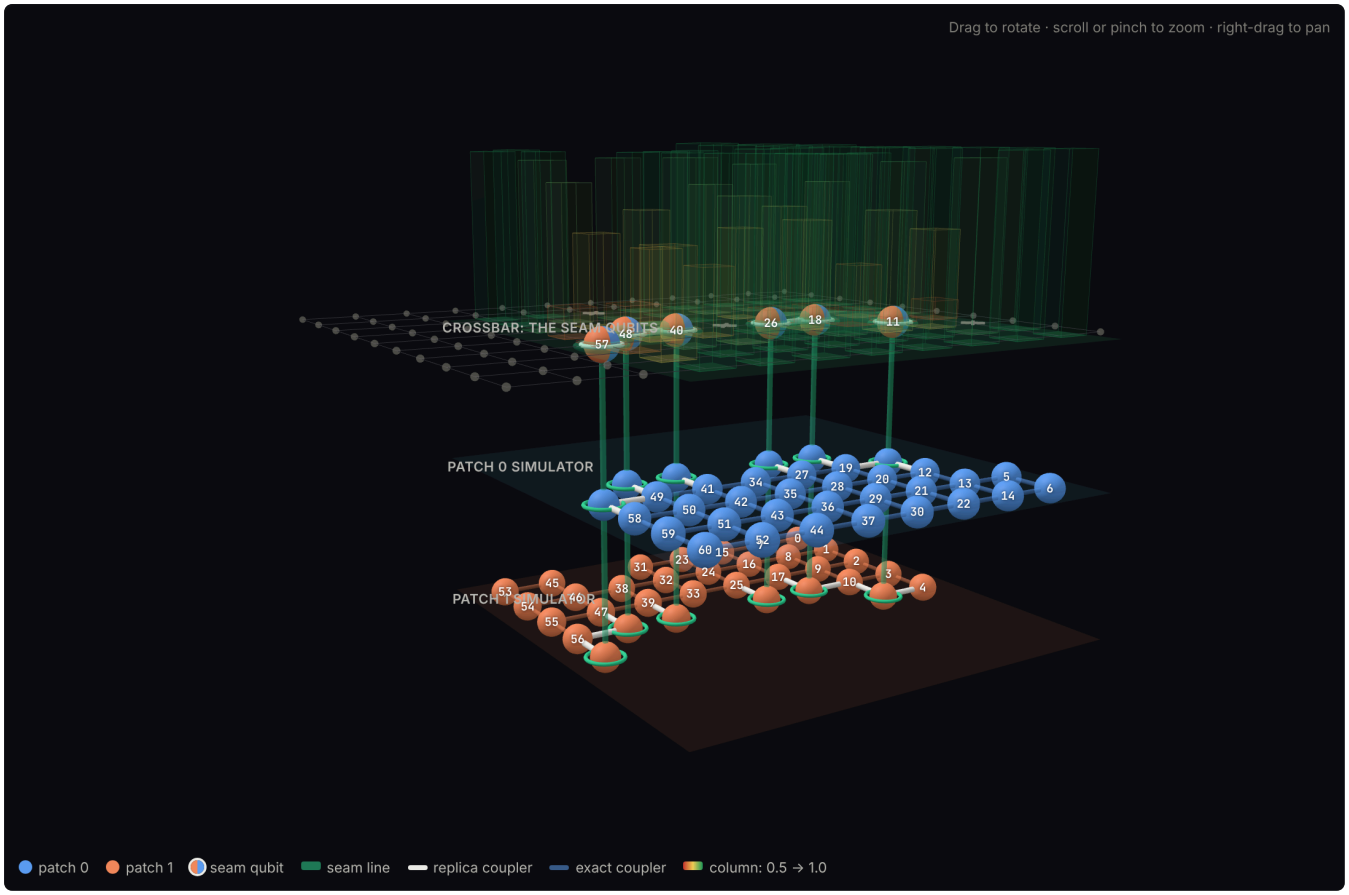


Figure 2 · nighthawk61r2 under the Walsh echo, patches pulled apart: the crossbar of seam qubits on top with F_GHZ columns, then the patch 0 and patch 1 simulators, joined by a pillar per seam qubit. The short columns sit over the seam.

The records are read by nighthawk_graph.py: views 18–21 draw the per-site fidelity map, the Layer 2 modes, ACE against the exact reference and the chip-wide break maps, and --grid3d writes one 3D page per register.

10. Scope of the hypotheses

Theorem 3 is exact under A1–A4 and says nothing outside them. The excluded effects are:

- Exchange coupling ($\eta > 0$). The Walsh sequence leaves $\eta \Sigma J_{ij} X_i X_j$ at first order (Remark on Theorem 1); §8 measures the cost at $\eta = 0.15$. A sequence mixing X and Y pulses is the natural extension.
- Finite-width or imperfect echo pulses (A1).
- Incoherent noise that is not global depolarising (A2). Theorem 2 uses unitary invariance of D_p; amplitude damping, local noise and leakage break it. Pauli twirling produces a Pauli channel, not D_p; global depolarising needs Clifford twirling or direct validation, for example a flat floor over the outcomes outside {0, 4, 8, 12}.
- Crosstalk during gates (A3).

- Finite sampling. Lemmas 3–5 are exact in expectation; finite shots and finite noise realisations add estimation error, measured in §8.
- Device values. The topology of Lemma 1 is the published one; the noise values in §8 are model parameters at a stress level, not device calibration data.
- Seam-partitioned simulation. On Qrack's ACE backend the seam acts as a channel on part of the register (Remark in §9.3), outside A2; Theorems 2 and 3 do not cover units that hold a seam qubit, and §9.2 measures refocusing lowering their fidelity.

References

- IBM Quantum Nighthawk processor announcement, Help Net Security, 12 Nov 2025: 120 qubits, 218 tunable couplers, square lattice.
- Reference implementation: shrs_nighthawk_stealth.py, numpy/scipy density-matrix simulation, seed 42.
- Chip-scale implementation: nighthawk_ghz.py (Layers 1a and 2 on ACE, with an exact QrackSimulator reference) and nighthawk_graph.py (views 18–21, 3D pages), shrs-nighthawk folder of twobombs/thereminq-examples; PyQrack 2.29.5, Qrack 10.13.2, seed 42.

Method literature, cited from memory and not re-opened for this document:

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- H. Hayes, K. Khodjasteh, L. Viola, M. J. Biercuk, Reducing sequencing complexity in dynamical quantum error suppression by Walsh modulation, Phys. Rev. A 84, 062323 (2011).
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- B. Swingle, Unscrambling the physics of out-of-time-order correlators, Nature Physics 14, 988 (2018).
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